MIL-S-63302(AR)
AMENDMENT 4
7 June 1983
SUPERSEDING
AMENDMENT 3
18 November 1982

MILITARY SPECIFICATION

SAFETY AND ARMING ASSEMBLIES
FOR
MINES: M74 AND M75

This Amendment forms a part of Military Specification MIL-S-63302(AR), dated 24 July 1979, and is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Change "XM74 and XM75" to "M74 and M75"; and wherever it appears in specification.

PAGE 2

MIL-L-63246: Delete "title" and substitute "Leads, Explosives, PA509, PA512, PA513, PA514, PA516, PA517 and PA521; Parts for, and Loading, Assembling and Packing".

PAGE 4

3.3, line 3: Delete "... of MIL-STD-331 except as follows: ... in 4.5.8.1." and substitute "... of MIL-STD-331 except as follows: Side output from two HMX loaded cups placed end to end shall produce .020 inch minimum (min) dent in an aluminum witness plate when tested as specified in 4.5.8.1. Side output from a single HMX loaded cup shall produce a 0.013 inch min dent."

3.5.b, line 2: Change "... slider and its explosive trains to the in-line ..." to "... slider to move to the in-line ...".

3.6.1, line 4: Delete "in the aluminum witness plate,".

FSC 1345

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Change table as follows:

TEMPERATURE	DEPTH OF DENT
-35° $-5^{\circ}F$	.005 inch min
$+75^{\circ}$ $\pm 10^{\circ}F$	.005 inch min
$+145^{\circ}$ $+5^{\circ}F$	.005 inch min

PAGE 5

3.6.2:

Line 5: Delete "... in a steel confining ring (witness block) ...".

Line 10: Delete "Dwg. 93287516" and substitute "Dwg. 9328756".

* 3.8: Delete "The resistance ... ohm." and substitute "The detonator(s) and micropiston actuator circuits shall be shunted to each other and to ground circuit (S&A housing)."

PAGE 9

TABLE I: Cable Flexible (Dwg. 9329826): Delete "Salt Spray and associated entries".

PAGE 12

* Delete "Spacer, Detonator .... 4.4.2.31" in its entirety.

PAGE 13

TABLE I:

Line 7: Delete "(Dwg. 9292756)" and subsitute "(Dwg. 9328756)".

Line 10: Delete "MFD" and substitute "MDF".

* Line 20: After "examination" add "(mechanical, electrical)"

* Add to requirement paragraph: ",3.8".

PAGE 14

TABLE I:

* Line 1: Delete in its entirety.

* Line 10: After "examination" add "(mechanical, electrical)".

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* Add to requirement paragraph: ",3.8".

* Line 15: Delete in its entirety.

Line 16, MDF TRAIN function, No. of Sample Units Column: Add "**".

Line 18, INSPECTION METHOD Column: Delete "4.5.18.2 and substitute "4.5.18.1".

Notes, line 2: Delete "Half the sample shall ..... facing down." and substitute "** Half of the sample units shall be functioned with lock plate facing up and the remaining sample units with lock plate facing down."

PAGE 16

4.4.2.1:

Major 105: Change "width" to "with".

Add "Major 112 - Length to main lead blast hole, 4 places - 0.40% - 3.2 - Gage".

Minor 202: Change "rang" to tang".

Minor 204: Delete in its entirety.

PAGE 17

4.4.2.2: Delete "Majors 104 and 105 in their entirety".

PAGE 18

4.4.2.3, In Note (a): Change "Inspection Level S-3" to "Sample Level S-1".

Major 106: Delete "ASTM-E-18" and substitute "ASTM-E-384".

PAGE 19

4.4.2.4:

Major 103: Change "Thickness of plate" to "Thickness of plate - detail A of drawing".

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Major 108: Delete "ASTM-E-18" and substitute "ASTM-E-18/ASTM-E-384".

Note (b): Change "Inspection Level S-3" to "Sample Level S-1".

PAGE 20

4.4.2.5: Delete Majors 121 and 122 in their entirety.

PAGE 23

4.4.2.7, Major 105: Change to read: "Chamfer missing, 2 places - 0.40% - 3.2 - Visual".

PAGE 24

4.4.2.8, Note (a): Change "Inspection Level S-3" to "Sample Level S-1".

Major 112: Delete "ASTM-E-18" and substitute "ASTM-E-384".

PAGE 25

4.4.2.9: Delete Major 101 in its entirety.

PAGE 28

4.4.2.12:

Major 112: Change to read: "True position of largest holes".
Major 114: Change to read: "True position of smallest hole with Datum A and B".

PAGE 30

4.4.2.14:

Major 119: Change to read: "Length of cutout for lock plate".

Add a second sentence to Notes: "The features may be gaged before the chromate finish is added provided that after parts are chromated they still meet requirements of drawing."

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PAGE 31

4.4.2.14, Major 126: Change to read: "Wall thickness at end of slider cavity".

PAGE 32

4.4.2.14:

Majors 140 and 141: Delete "DET. L " and substitute "DET. M".

* Major 142: Delete "Gage" and substitute "Visual".

PAGE 33

4.4.2.15:

Major 105: Delete "Sec G-G" and substitute "Detail C".

Add a second sentence to Notes: "The features may be gaged before the chromate finish is added provided that after parts are chromated they still meet requirements of drawing."

PAGE 38

4.4.2.18:

Major 103: Change "imcomplete" to "incomplete".

Major 106: Change to read: "Length of roll crimped cup, min, 2 places".

Major 109: Change to read: "Spacing present between MDF and HMX, 2 places - 100% - 3.2 - 4.5.11".

Add "Major 112: Inclusions, voids or impurities - 100% - 3.2 - 4.5.11".

Add this sentence to Note (a): "The samples from Major 107 may be used in output test of Major 110."

Add Note (b): "Results from #110 shall be recorded and forwarded to DRDAR-QAT-M (See 6.5)."

PAGE 39

4.4.2.19:

Majors 102, 103, 105, 108: Delete "GAGE" and substitute "Visual".

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Major 107: Delete "True position of groove pin to datum A & B" and substitute "True position of tabs to datum A & B, 2 places".

PAGE 40

4.4.2.20: Delete Major 103 in its entirety.

PAGE 42

4.4.2.22, Major 113: Change to read: "Height between tabs".

PAGE 43

* 4.4.2.22, Major 121: Delete "clearace" and substitute "clearance".

PAGE 44

4.4.2.23, Major 101: Change to read: "Explosive lead, PA517, inverted, 2 places".

PAGE 45

4.4.2.24:

Major 102: Change to read: "Explosive lead, PA517, inverted, 2 places".

Note (a): Change "Inspection Level I" to "Sample Level S-3".

Delete last sentence of Note (a): "Any failure ..... to this test."

PAGE 47

4.4.2.26: Delete Major 103 in its entirety.

PAGE 52

* Delete para. 4.4.2.31 in its entirety.

PAGE 53

4.4.2.32: Add "Major 109 - Length of smallest diameter - 0.40% - 3.2 - Gage".

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4.4.2.36, Notes: Delete "*See 4.5.2" and substitute "*See 4.5.3".

PAGE 58

4.4.2.37:

Change title to read: "Safety and Arming Assembly (Prior to assembling Upper and Lower Housing Assemblies)".

Critical 2: Delete in its entirety.

Critical 5: Delete in its entirety.

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4.4.2.38:

Change title to read: "Safety and Arming Assembly (Prior to assembling Upper and Lower Housing Assemblies)".

Critical 2: Delete in its entirety.

Critical 5: Delete in its entirety.

PAGE 60

4.4.2.39, Critical 1: Change "3.5.7.2" to "4.5.7.2".

PAGE 61

4.4.2.40, Critical 1: Change "3.5.7.2" to "4.5.7.2".

PAGE 62

4.4.2.41, Major 101: Change to read: "Actuator spacer, blast shield assy and flexible cable assy missing or incorrectly installed".

PAGE 63

4.4.2.42, Major 103: Change to read: "Blast shield assy, actuator spacer, MDF spacer and flexible cable assy missing or incorrectly installed".

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4.4.2.43:

* Critical 1: Add to requirement paragraph ",3.8".

* Critical 2: Change to read: "Length to slider assy, over max.".

Critical 3: Change to read: "Slider release pin and shear pin missing or improperly assembled".

Add Critical 4: Centrifugal lock binding, sticking or restricted in travel to armed position and/or in return to the safe position, 2 places, (c), 100%, 3.2, Visual/Manual

Major 102: Delete "Gage" and substitute "Gage/Visual".

Add Note (c): Prior to installing lock plate, each centrifugal lock must be moved to the fully armed position and observation made of free return of each lock to safe position.

Major 109: Delete "Electrical Continuity and Impedance Test" and substitute "Electrical Resistance Test".

4.4.2.44:

Critical 1: Change to read: "Eyelet ground circuit is open, detonators or micro piston actuator not commonly shorted to S&A body".

* Requirement paragraph: Add ",3.8".

* Critical 2: Change to read: "Length to slider assy, over max.".

Critical 3: Change to read: "Slider release pin and shear pin missing or improperly assembled".

Add Critical 4: "Centrifugal lock binding, sticking or restricted in travel to armed position and/or in return to the safe position, 2 places, (c), 100%, 3.2, Visual/Manual".

Major 102: Delete "Gage" and substitute "Gage/Visual".

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Major 110: Delete "Electrical Continuity and Impendance Test" and substitute "Electrical Resistance Test".

Add Note (c): "Prior to installing lock plate, each centrifugal lock must be moved to the fully armed position and observation made of free return of each lock to safe position."

PAGE 66

4.4.2.45:

* Line 3, under "Examination or Test": Delete "Transportation vibration" and substitute "TRANSPORTATION VIBRATION".

* Line 4, Following examination: Add "(mechanical, electrical shorting)".

* Line 5: Delete "Continuity resistance checks, -, -, 4.5.14".

* Add Line 6: "Electrical resistance tests, Note 4, -, 3.2, 4.5.14".

Line 8, Final arming: Delete "4.5.7.2" and substitute "4.5.17.2".

Add Note 3: "Results from in-line explosive functioning shall be recorded and forwarded to DRDAR-QAT-M (See 6.5)."

* Add Note 4: "Must be performed following T.V. and before in-line functioning".

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4.4.2.46:

* Line 4, Following examination: Add "(mechanical, electrical shorting)".

* Line 5: Delete "Continuity resistance checks, -, -, 4.5.14".

* Add Line 6: "Electrical resistance tests, Note 4, -, 3.2, 4.5.14".

Line 11, Main charge train functioning: Delete "4.5.18.2" and substitute "4.5.18.1".

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Add Note 3 : "Results from in-line explosive functioning shall be recorded and forwarded to DRDAR-QAT-M (See 6.5)."

* Add Note 4: "Must be performed following T.V. and before in-line functioning".

PAGE 73

4.5.8.1:

Line 6: Change "Witness plate, 5/8 X 1 1/4 X 1 1/4 inches,..." to "Witness plates, 5/8 X 2 X 18 inch for series end to end output and 1/8 X 1/2 X 3/4 inches for single cup output, ...".

Add to end of paragraph: "A maximum of ten (10) test samples may be tested end to end at one time on a single 5/8 X 2 X 18 inch witness plate. MDF shall be formed at a 90° +5° angle with a .047 inch inside bend radius."

4.5.8.2, line 5 (last sentence): Change to read: "Acceptable assemblies subjected to this test may be used in output test of 4.5.8.1."

4.5.9, line 5 (last sentence): Change to read: "This is a non-destructive test and acceptable parts meeting no-movement requirement of drawing may be returned to the lot."

4.5.11, 4.5.11.1 and 4.5.11.2: Delete in their entirety and substitute: "4.5.11 Neutron radiographic examination. The MDF assembly, Dwg. 9328723, shall be subjected to neutron radiographic examination. Observation shall be made for compliance with requirements of the drawing. This is a non-destructive test and acceptable parts may be returned to the lot."

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4.5.13, change last sentence to read: "This is a non-destructive test and acceptable parts may be returned to the lot."

4.5.15.3: Add the following: "NOTE: The S&A assembly may be tested in standard shipping (service item) package in lieu of rigid mounting in a fixture."

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* 4.5.14: Delete "Following ... (Non-destructive test)" and substitute "Following removal of eyelet after transportation vibration or after assembly of the flexible cable to appropriate S&A and the drilling of a hole for the subsequent insertion of a shorting eyelet, the flexible cable assembly shall be subjected to the resistance tests specified on applicable S&A drawing. Readout meter shall be a milliohmeter or equivalent with a test current of one milliampere, or ten milliamperes, limited to ten seconds max. Observation shall be made for compliance with requirements of the applicable drawing. After resistance measurements, the electro explosive devices shall have their input conductors positively shorted and tested for shorting requirement specified in 3.8 and applicable drawing. (Non-destructive test)"

PAGE 75

4.5.17.1, Line 1: Delete "The S&A test ... about its center line,... " and substitute "The S&A test samples from 4.5.16 shall be placed in an approved fixture and spin machine, and subjected to a spin of 2500 RPM max about its center line, ...".

4.5.17.2, line 2: Delete "form" and substitute "from".

4.5.17.2.a: Delete in its entirety and substitute: "An aluminum test plate, 1/2 X 3/4 X 1/8 inch thick, shall be secured at the side face of the MDF cap preselected for functioning. Material for the aluminum witness plate shall be as specified in 4.5.8.1 (Dwg. 9328756 only)."

4.5.17.2.b: Delete in its entirety and substitute: "The S&A assembly under test shall be inclosed by four (4) steel test plates located at the output of each of the four (4) main explosive leads. Each test plate shall be approximately 1 1/8 X 1 1/8 X 5/8 inch of cold drawn or cold rolled steel, per ASTM-A-108, C1018 or C1020, Rockwell hardness B80 - B95. As an alternate, a cylinder of the above material may be used. The cylinder with an outside diameter of 2.00 inches and an inside diameter of 1.168 + .005 inches with a height sufficient to contain the output shall be secured to preclude displacement during test."

PAGE 76

4.5.17.2.f, line 2: Delete "... the energy from a one hundred ... volts max." and substitute "... the energy from an eighty-two (82) $\pm$ 5% microfarad capacitor charged to five (5) $\pm$ 0.1% volts d.c.."

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4.5.18.1: Delete in its entirety and substitute "The S&A assemblies previously armed and fitted with witness plates, see 4.5.17.2.a, b, shall be functioned by subjecting the parallel M100 and PA506 delay detonators to the energy from a 82 ± 5% microfarad capacitor charged to 5.0 ± 0.1% volts d.c. Observation shall be made for compliance with requirements of 3.6.1 and 3.6.2. The 20 ms to 50 ms delay time of the delay detonator PA506 shall be recorded".

4.5.18.2: Delete in its entirety and substitute "The S&A assemblies previously armed and fitted with steel witness plates, see 4.5.17.2.b, shall be functioned by subjecting the M100 detonator to the energy from an eighty-two (82) ± 5% microfarad capacitor charged to five (5) ± 0.1% volts d.c. Observation shall be made for compliance with 3.6.2."

* 4.5.19, line 10: Following "... of slider two lock tabs." add "Verification shall be made for positive shorting of the detonator(s) and micropiston actuator as per requirement of 3.8 and applicable drawing."

PAGE 77

6.5: Delete in its entirety and substitute "Submission of ammunition data cards and performance data. In addition to the normal distribution of records, one copy of ammunition data cards, in-line explosive performance data (4.4.2.45, 4.4.2.46) and MDF assembly output test (4.4.2.18) shall be forwarded to Commander, ARRADCOM, ATTN: DRDAR-QAT-M, Dover, NJ 07801."

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